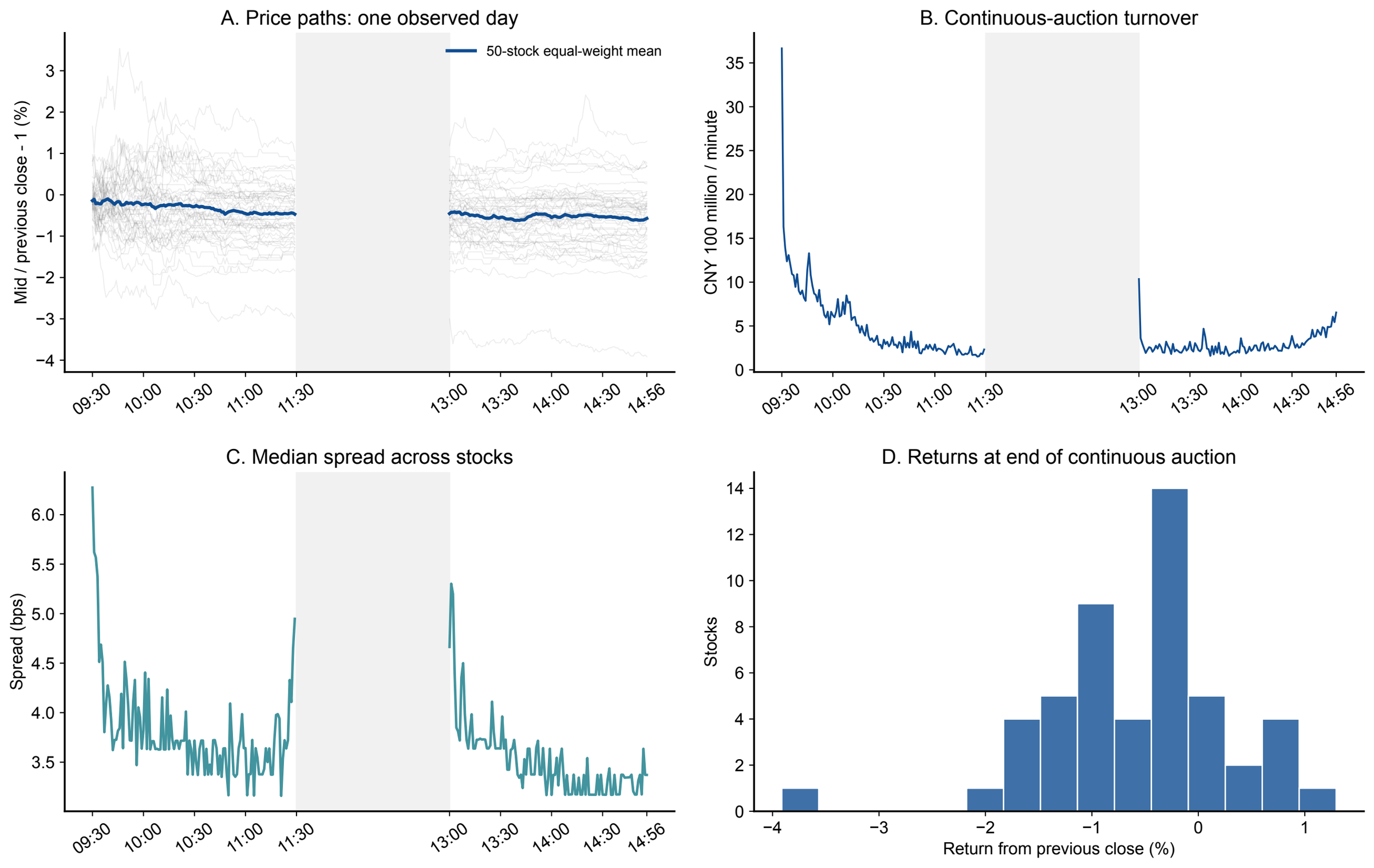
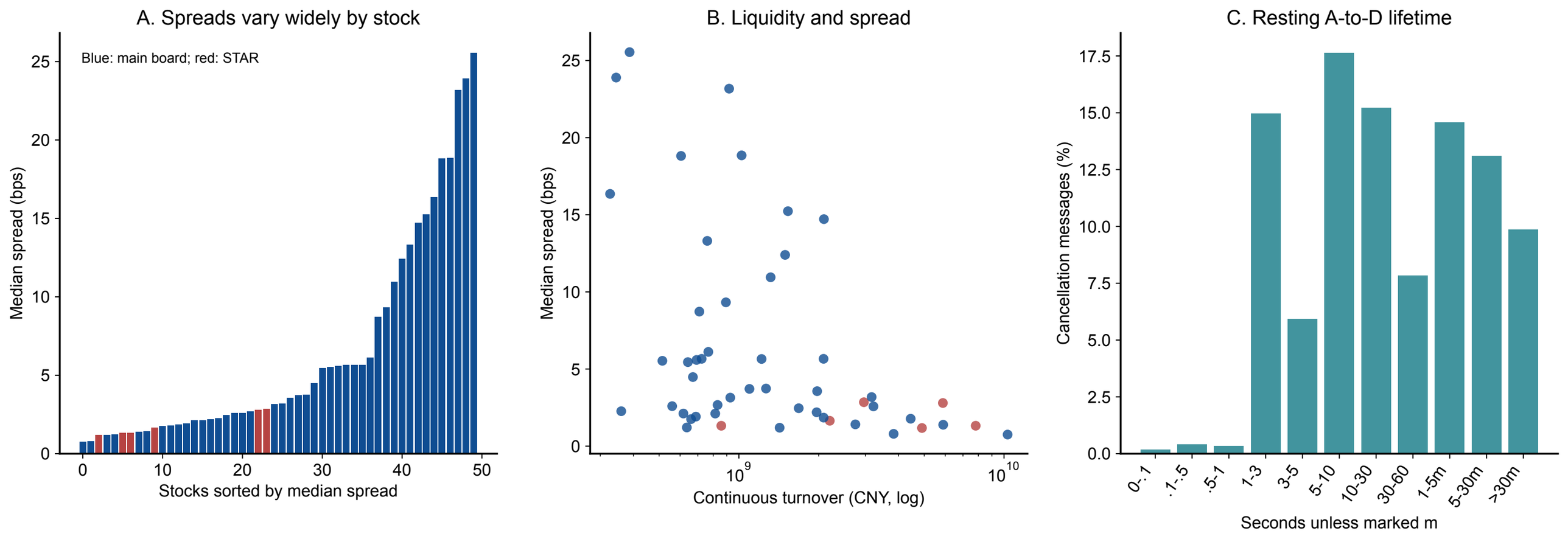
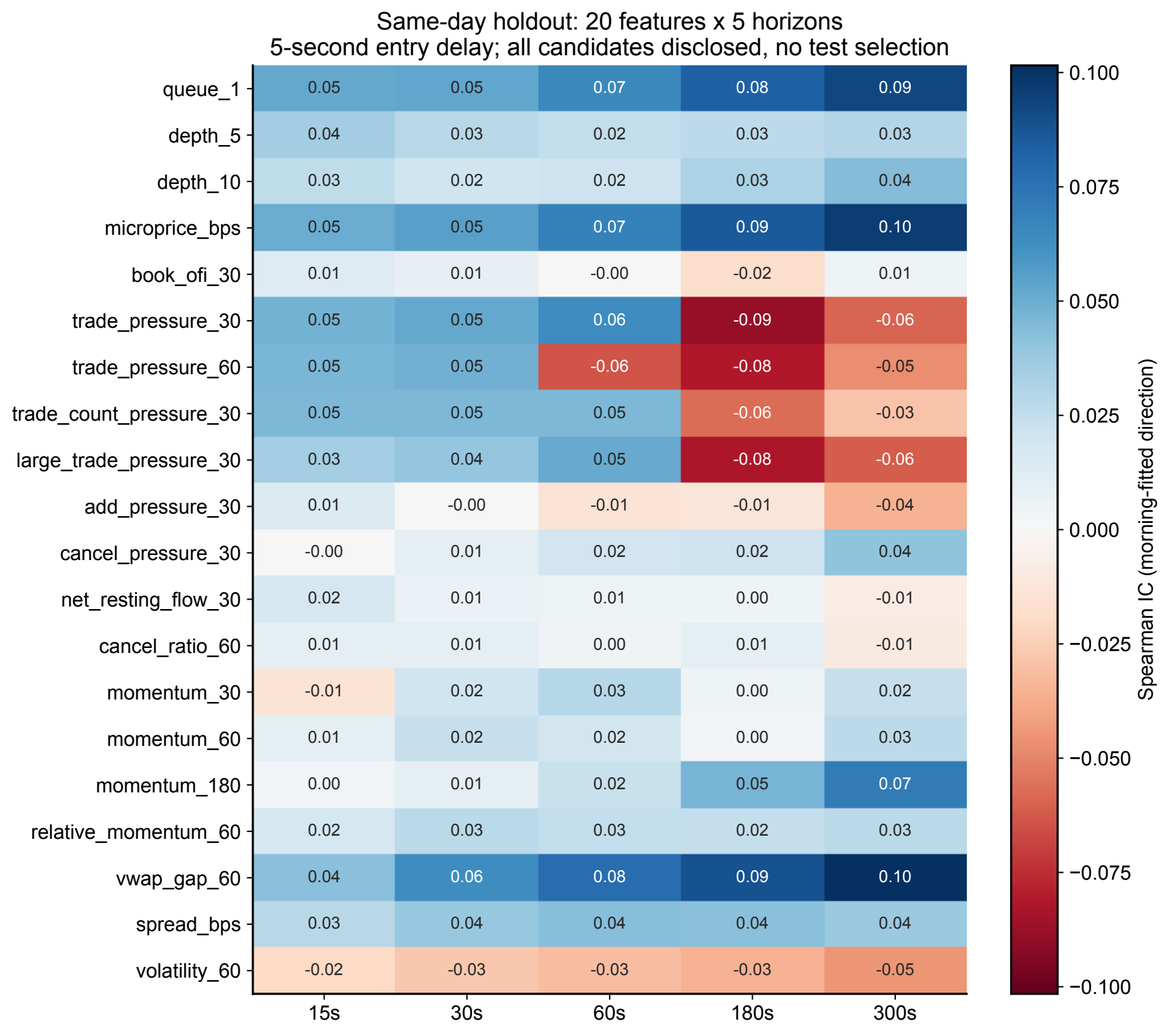
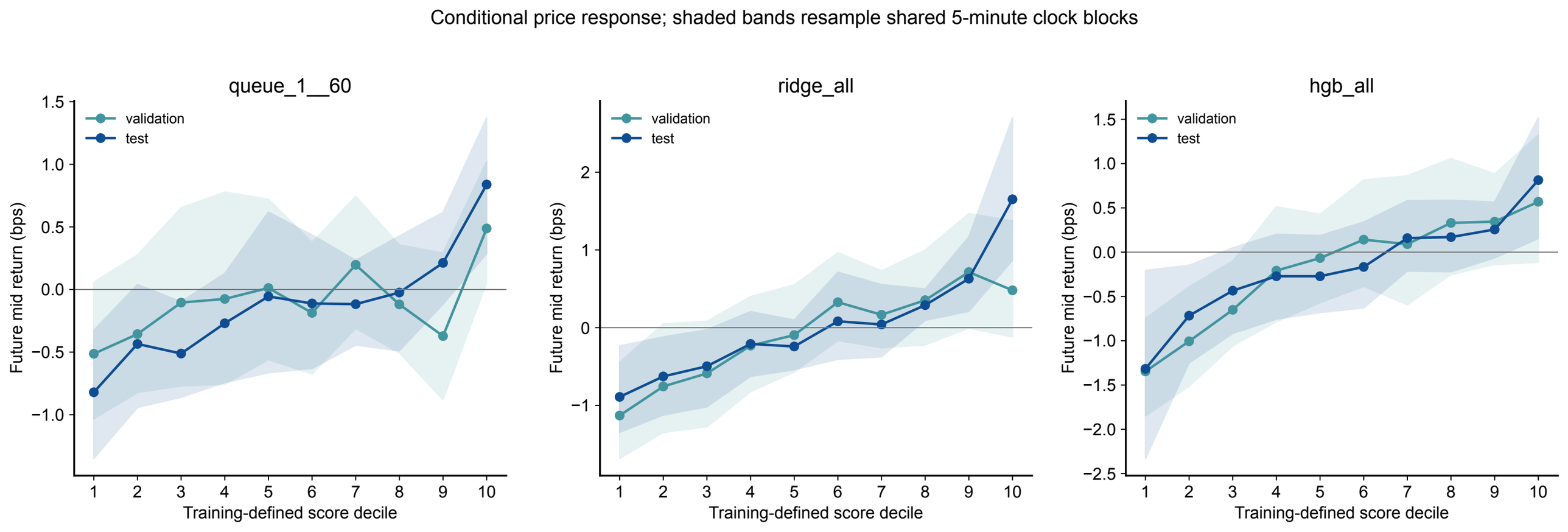
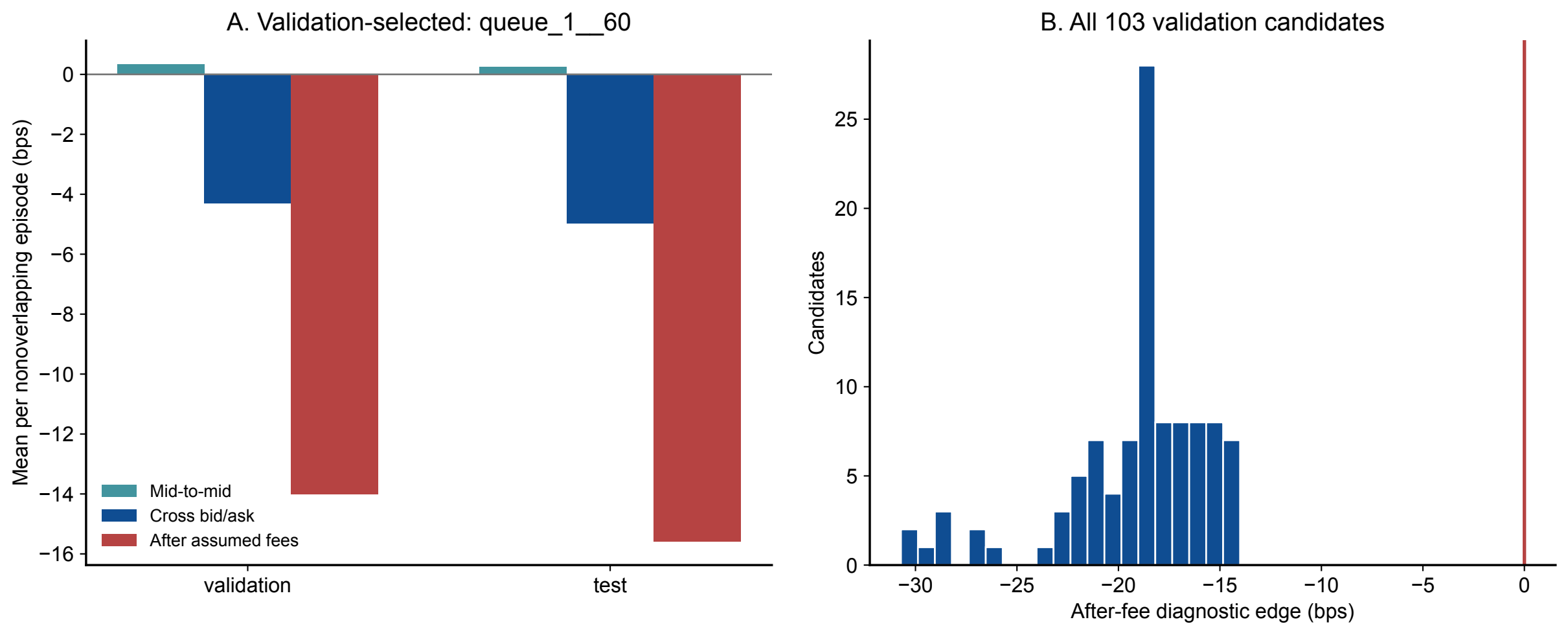
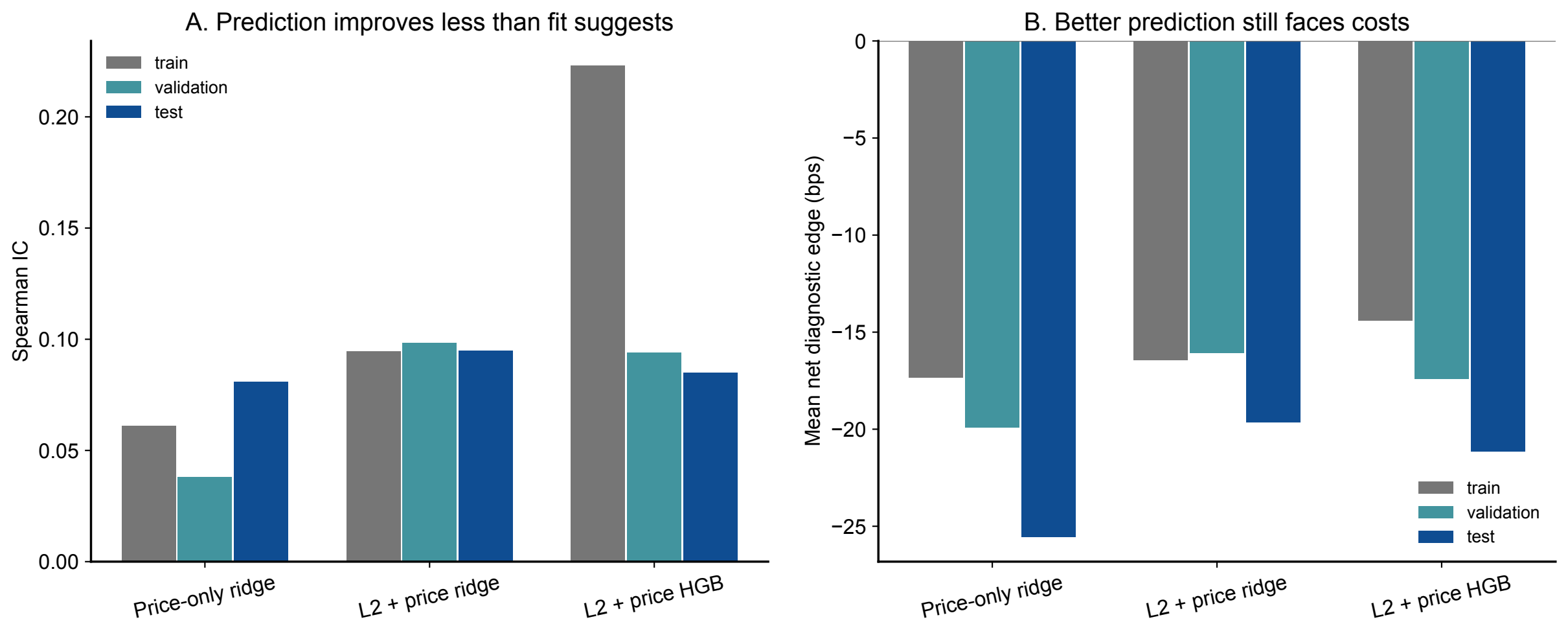

A. Price paths: one observed day
50-stock equal-weight mean
Mid / previous close - 1 (%)
3
2
1
0
−1
−2
−3
−4
09:30
10:00
10:30
11:00
11:30
13:00
13:30
14:00
14:30
14:56
B. Continuous-auction turnover
CNY 100 million / minute
35
30
25
20
15
10
5
0
C. Median spread across stocks
Spread (bps)
6.0
5.5
5.0
4.5
4.0
3.5
D. Returns at end of continuous auction
Stocks
14
12
10
8
6
4
2
0
−4
−3
−2
−1
0
1
Return from previous close (%)

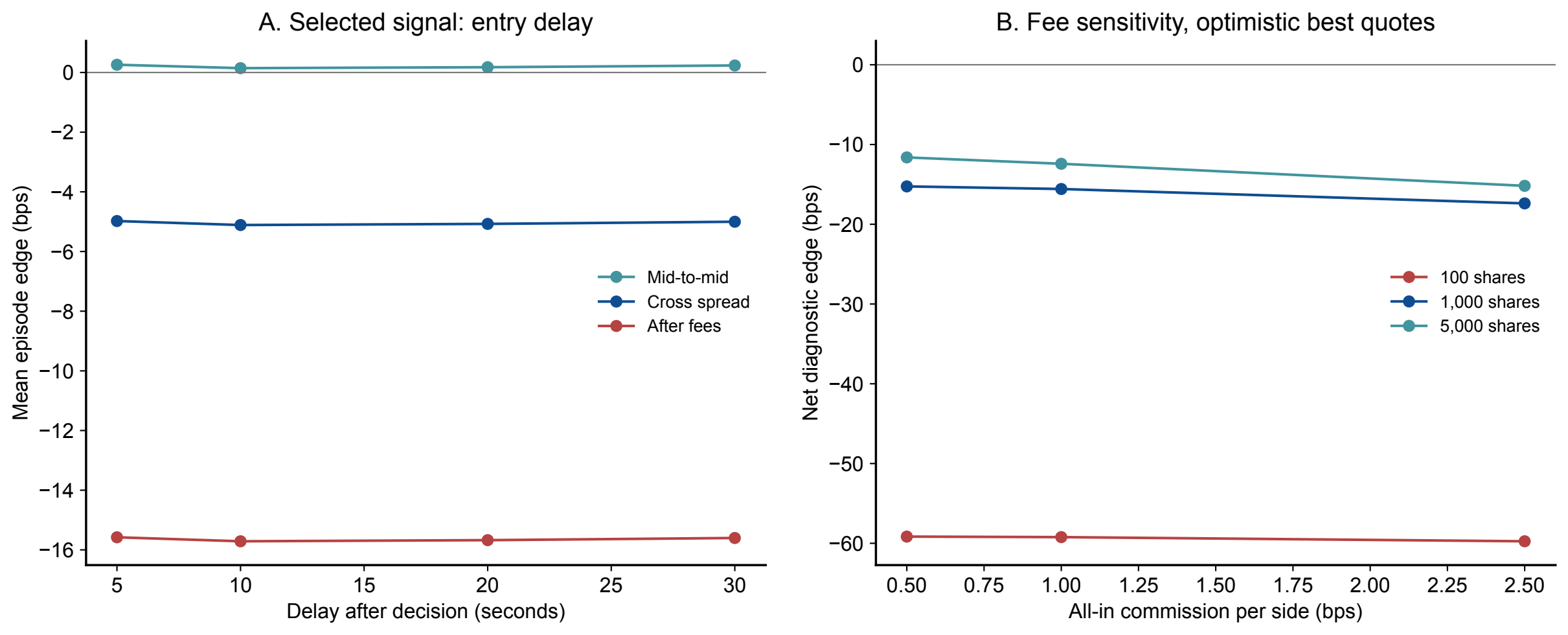

A. Spreads vary widely by stock
Blue: main board; red: STAR
Median spread (bps)
Stocks sorted by median spread
B. Liquidity and spread
Median spread (bps)
Continuous turnover (CNY, log)
C. Resting A-to-D lifetime
Cancellation messages (%)
0-.1
.1-.5
.5-1
1-3
3-5
5-10
10-30
30-60
1-5m
5-30m
>30m
Seconds unless marked m


Same-day holdout: 20 features x 5 horizons
5-second entry delay; all candidates disclosed, no test selection
queue_1 0.05 0.05 0.07 0.08 0.09
depth_5 0.04 0.03 0.02 0.03 0.03
depth_10 0.03 0.02 0.02 0.03 0.04
microprice_bps 0.05 0.05 0.07 0.09 0.10
book_ofi_30 0.01 0.01 -0.00 -0.02 0.01
trade_pressure_30 0.05 0.05 0.06 -0.09 -0.06
trade_pressure_60 0.05 0.05 -0.06 -0.08 -0.05
trade_count_pressure_30 0.05 0.05 0.05 -0.06 -0.03
large_trade_pressure_30 0.03 0.04 0.05 -0.08 -0.06
add_pressure_30 0.01 -0.00 -0.01 -0.01 -0.04
cancel_pressure_30 -0.00 0.01 0.02 0.02 0.04
net_resting_flow_30 0.02 0.01 0.01 0.00 -0.01
cancel_ratio_60 0.01 0.01 0.00 0.01 -0.01
momentum_30 -0.01 0.02 0.03 0.00 0.02
momentum_60 0.01 0.02 0.02 0.00 0.03
momentum_180 0.00 0.01 0.02 0.05 0.07
relative_momentum_60 0.02 0.03 0.03 0.02 0.03
vwap_gap_60 0.04 0.06 0.08 0.09 0.10
spread_bps 0.03 0.04 0.04 0.04 0.04
volatility_60 -0.02 -0.03 -0.03 -0.03 -0.05
15s
30s
60s
180s
300s
Spearman IC (morning-fitted direction)
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100

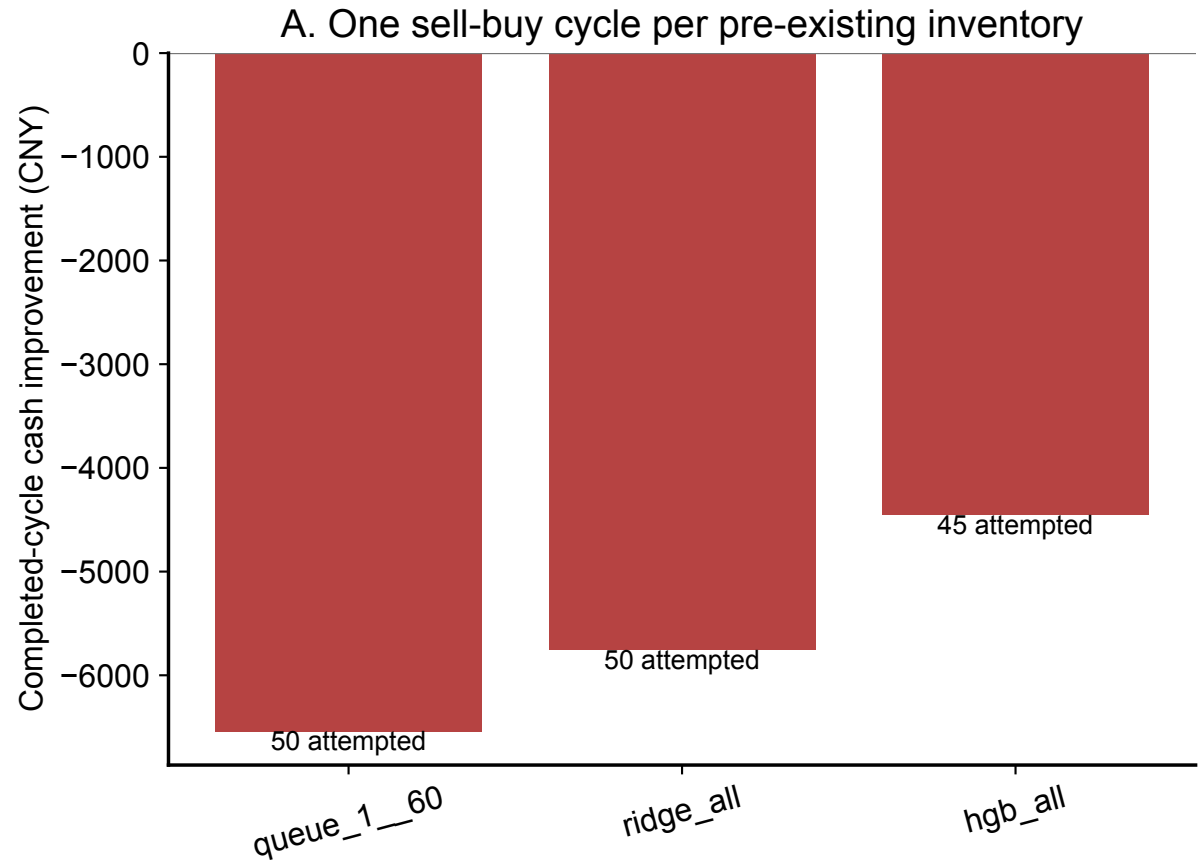

Conditional price response; shaded bands resample shared 5-minute clock blocks
queue_1__60
ridge_all
hgb_all
validation
test
Future mid return (bps)
Training-defined score decile

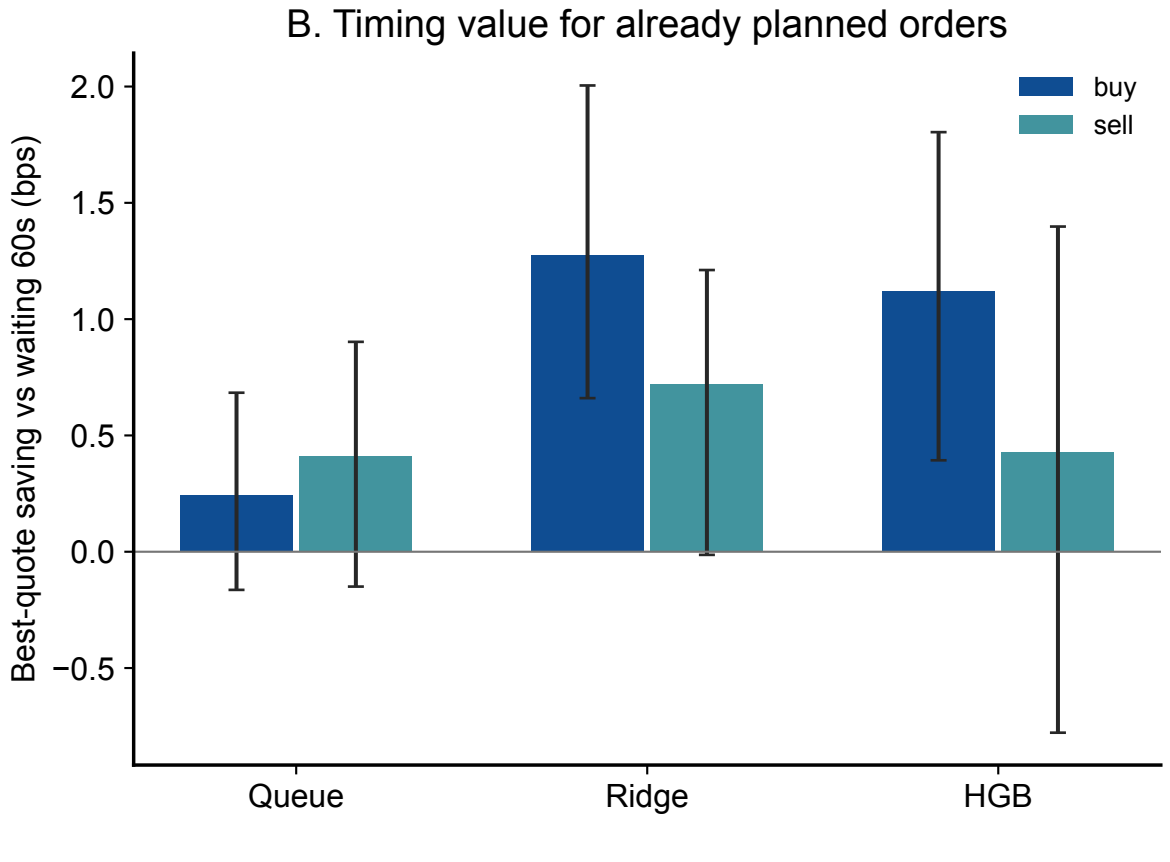

A. Validation-selected: queue_1__60
Mean per nonoverlapping episode (bps)
0
−2
−4
−6
−8
−10
−12
−14
−16
Mid-to-mid
Cross bid/ask
After assumed fees
validation
test
B. All 103 validation candidates
Candidates
25
20
15
10
5
0
−30
−25
−20
−15
−10
−5
0
After-fee diagnostic edge (bps)

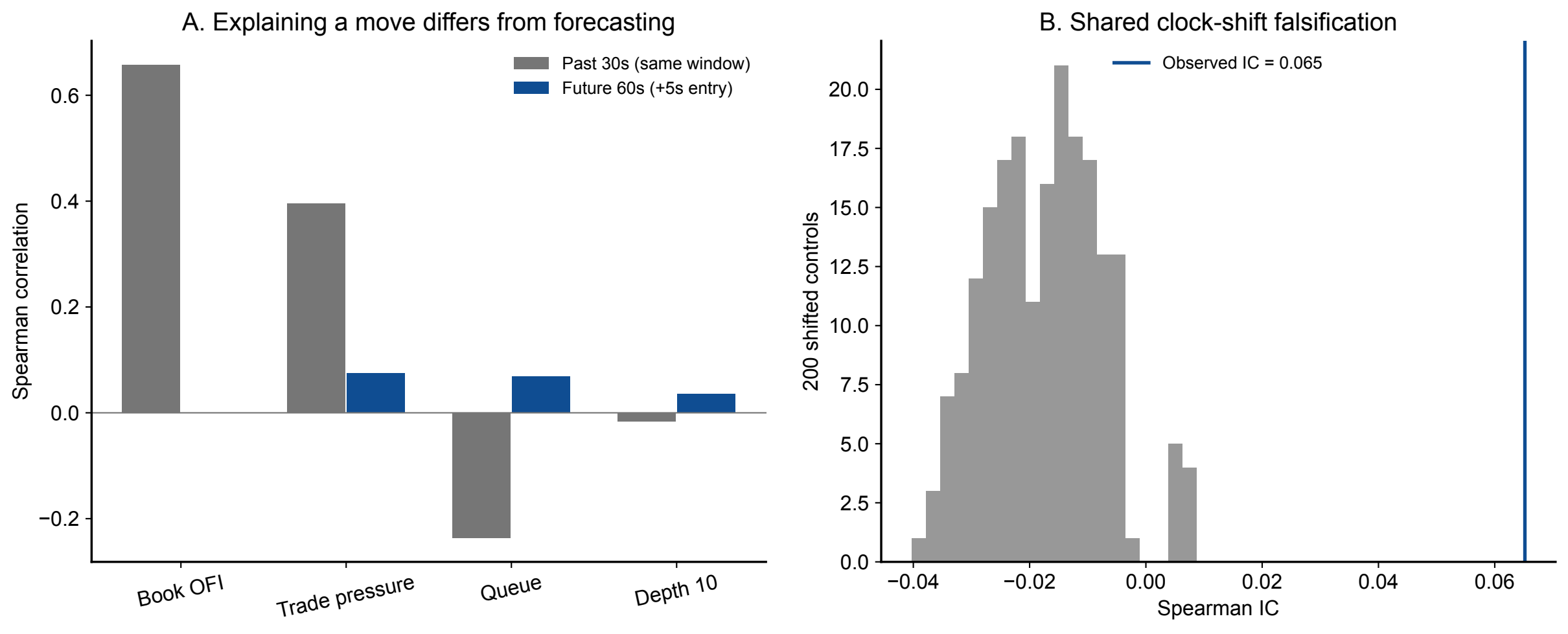

A. Prediction improves less than fit suggests
train
validation
test
Spearman IC
0.00
0.05
0.10
0.15
0.20
Price-only ridge
L2 + price ridge
L2 + price HGB
B. Better prediction still faces costs
Mean net diagnostic edge (bps)
0
−5
−10
−15
−20
−25
train
validation
test
Price-only ridge
L2 + price ridge
L2 + price HGB


A. Selected signal: entry delay
Mean episode edge (bps)
0
−2
−4
−6
−8
−10
−12
−14
−16
Mid-to-mid
Cross spread
After fees
5
10
15
20
25
30
Delay after decision (seconds)
B. Fee sensitivity, optimistic best quotes
Net diagnostic edge (bps)
0
−10
−20
−30
−40
−50
−60
100 shares
1,000 shares
5,000 shares
0.50
0.75
1.00
1.25
1.50
1.75
2.00
2.25
2.50
All-in commission per side (bps)

Inventory cycles use 10-level 1,000-share VWAP; timing panel is a quote benchmark, not realized PnL






A. Explaining a move differs from forecasting
Past 30s (same window)
Future 60s (+5s entry)
Spearman correlation
0.6
0.4
0.2
0.0
−0.2
Book OFI
Trade pressure
Queue
Depth 10
B. Shared clock-shift falsification
Observed IC = 0.065
200 shifted controls
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
−0.04
−0.02
0.00
0.02
0.04
0.06
Spearman IC